

A full-page background image of a triathlete, Ben Hill, running on a paved path. He is wearing a white triathlon jersey with yellow and green accents, black shorts, and a race bib. The background is a soft-focus view of a coastal area with green grass and a blue sky.

Ready for the long grind:

Australian triathlete Ben Hill competes in the 2025 Ironman event in Cairns. By understanding work and energy concepts in sports, athletes like him can improve their speed, endurance, power, and accuracy while also preventing fatigue and injuries.

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In sports, work and energy are involved in every movement and action. To explain in simple physics: in fast bowling, you transform chemical energy from your food into kinetic energy of your body while running up, creating momentum. The muscles store elastic potential energy to be converted into kinetic energy at the right moment. Part of this kinetic energy becomes elastic potential energy as the body positions itself, and gravitational potential energy as you rise above the ground. Then, this potential energy becomes kinetic energy while releasing the ball. The bowler aims to maximise the impulse (force multiplied by time) at the point of release.

The law of conservation of energy helps in efficiently conserving, transferring, and converting energy from one form to another, while accounting for energy lost due to friction, air resistance, heat and sound. This allows you to do more work and have more energy in sports. For example, when you swing a golf club, you want to uncork your wrists at the right angle so that the clubhead hits the ball with the greatest speed possible.

By understanding work and energy concepts in sports, one can improve speed, endurance, power, and accuracy while also preventing fatigue and injuries by managing energy levels and avoiding overexertion.

The science of motion

Another indispensable physics concept in sports is motion. Motion is the change of position or orientation of an object over time. It can be described by various quantities, such as speed, velocity, acceleration, momentum, and angular momentum. These quantities help you analyse and

predict the behaviour of moving objects in sports across different frames of reference.

An example of the synergy between angular momentum and rotational inertia is skating. When you skate, you use angular momentum to spin faster or slower. Angular momentum is the product of rotational speed and rotational inertia. It is sustained in an isolated system, which means it stays uniform unless acted upon by an external force.

This means: if you decrease your rotational inertia by pulling your arms and legs closer to your body, you will increase your rotational speed and spin faster. Conversely, if you increase your rotational inertia by extending your arms and legs, you will decrease your rotational speed and spin slower.

By mastering motion in sports, you can control and manipulate your own movement and that of other objects. You can also anticipate and react to the movement of your opponents and the environment. This gives you an advantage in reflex, speed, agility, stability, balance, and coordination.

The magnificence of the physics and sports combo

Physics and sports are intimately linked. Physics helps you understand how things work and why they happen in sport.

It also helps improve your performance, prevent injuries, and gain an edge over your opponents. It does not mean one has to be a scientist to understand human movement, angles and forces. A simple understanding of it can pave the way for excellence.

But physics is not only useful for sports but also understanding oneself, especially through the lens of quantum mechanics.